

The background is a vibrant, futuristic digital landscape. It features a perspective view of a floor made of a grid of squares, receding into the distance. The floor is illuminated with a mix of pink and blue light. On either side of the floor, there are jagged, low-poly structures that resemble digital mountains or data blocks. In the center of the image, a large, glowing pink circle is superimposed over a dark, starry space. The overall color palette is dominated by deep blues, purples, and bright pinks.

Computer Building 2023

Parts, wiring, and how it all goes together.

Parts List – Main Components

CPU – Central Processing Unit

Storage – HDD (Hard Drive Disk) or SSD (Solid State Disk)

RAM – Random Accessible Memory

Mother Board

PSU – Power Supply Unit

GPU – Graphic Processing Unit

CPU Cooler

The Case

Peripherals – Mouse/Keyboard/Monitor/Speakers

CPU History

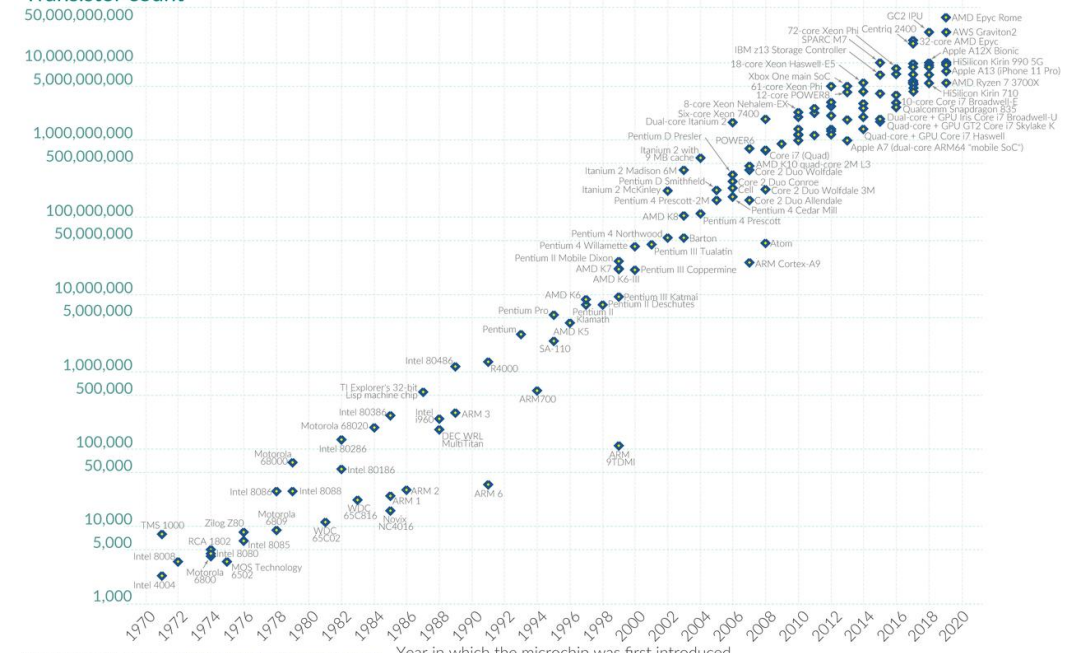
- + Binary Computing
- + How CPU's are made
- + Moore's Law

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

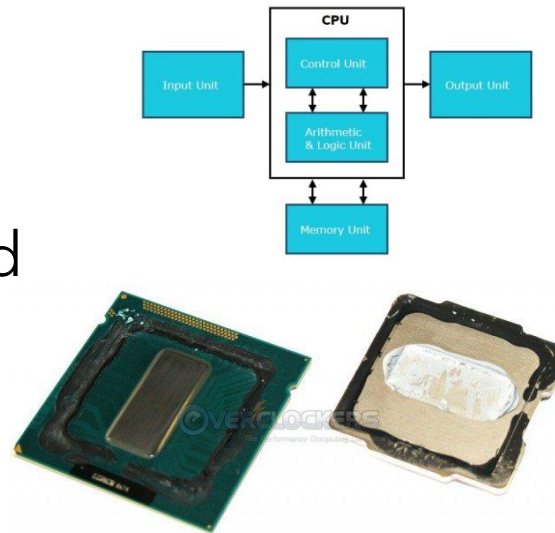
Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.



CPU

Intel vs AMD

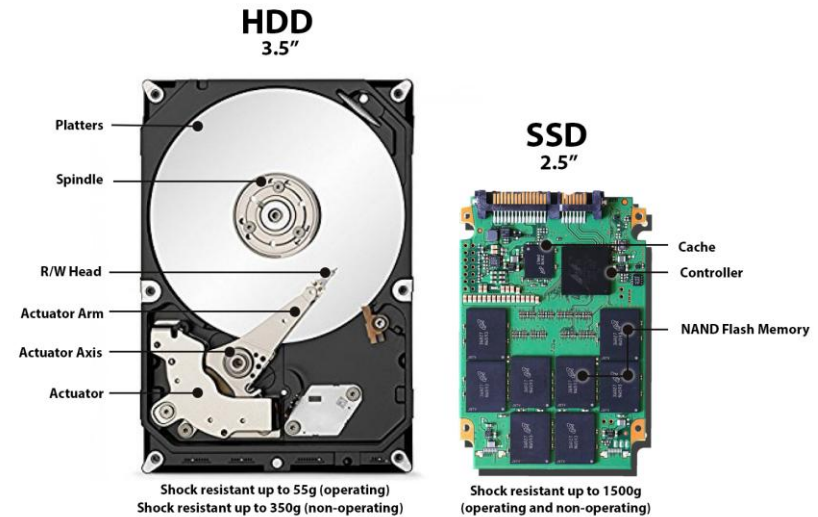
- + Core Count
- + Threading
- + Clock Speed
- + Socket Type



- + Control Unit (CU)
- + Arithmetic Logic unit (ALU)
- + Registers
- + Cache
- + Buses
- + Clock

Data Storage – HDD/SSD/SAS

- | | |
|--------------|---------------------|
| + HDD | + SSD |
| + RPM | + Read/Write Cycles |
| + Capacity | + Capacity |
| + Cache | + Cache |
| + Connection | + Connection |

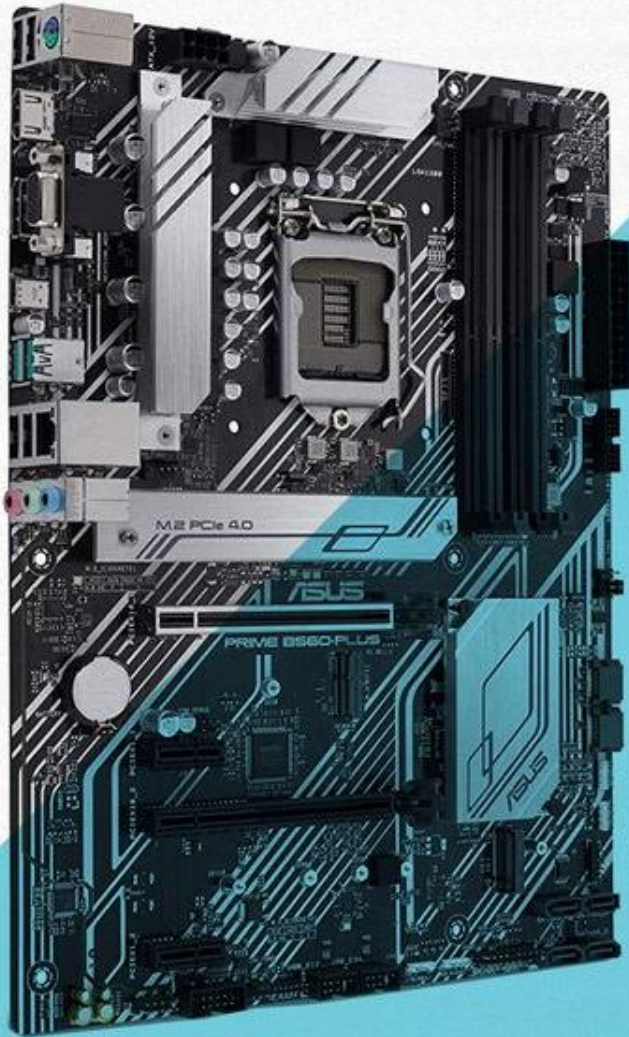


RAM

- + DDR - Double Data Rate
- + DIMM vs SODIMM
- + DDR4 vs DDR5
- + CAS
- + Latency



ComputerHope.com



Motherboard

The heart of it all

- + Feature Set
- + Network Card
- + Sound Card
- + I/O (Input/Output)
- + Compatibility
- + Growth
- + Size (ATX/Micro/EATX)

PSU – Power Supply Unit

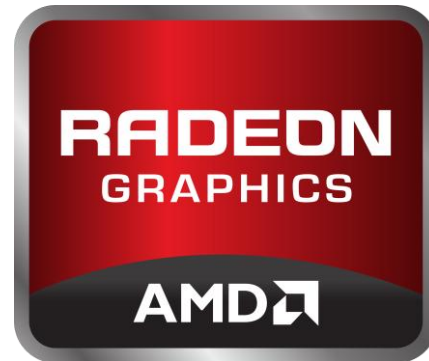
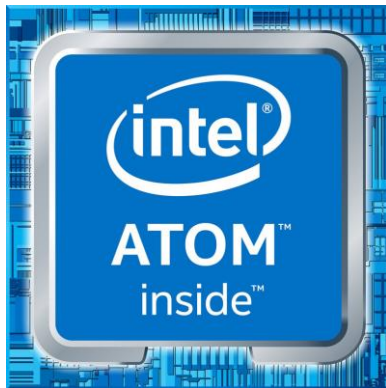
- + Rating
- + Modularity
- + Cooling
- + Capacity



GPU

Make it pretty on screen

- + Use Case
- + Red or Green
- + Compatibility
- + RGB (Not the LEDS)



CPU Cooling

Air VS Water

- + Compatibility
- + Air vs Water
- + AIO - All In One
- + Custom Loop
- + Air





The Case

- + Compatibility
- + Size
- + Rebuild Ability
- + Ease of Build
- + Front I/O



Peripherals

- + Monitor
- + Mouse
- + Keyboard
- + Speakers
- + Mic
- + Web Cam
- + Other



Parts List

Q/A

CPU – Central Processing Unit

Storage – HDD (Hard Drive Disk) or SSD (Solid State Disk)

RAM – Random Accessible Memory

Mother Board

PSU – Power Supply Unit

GPU – Graphic Processing Unit

CPU Cooler

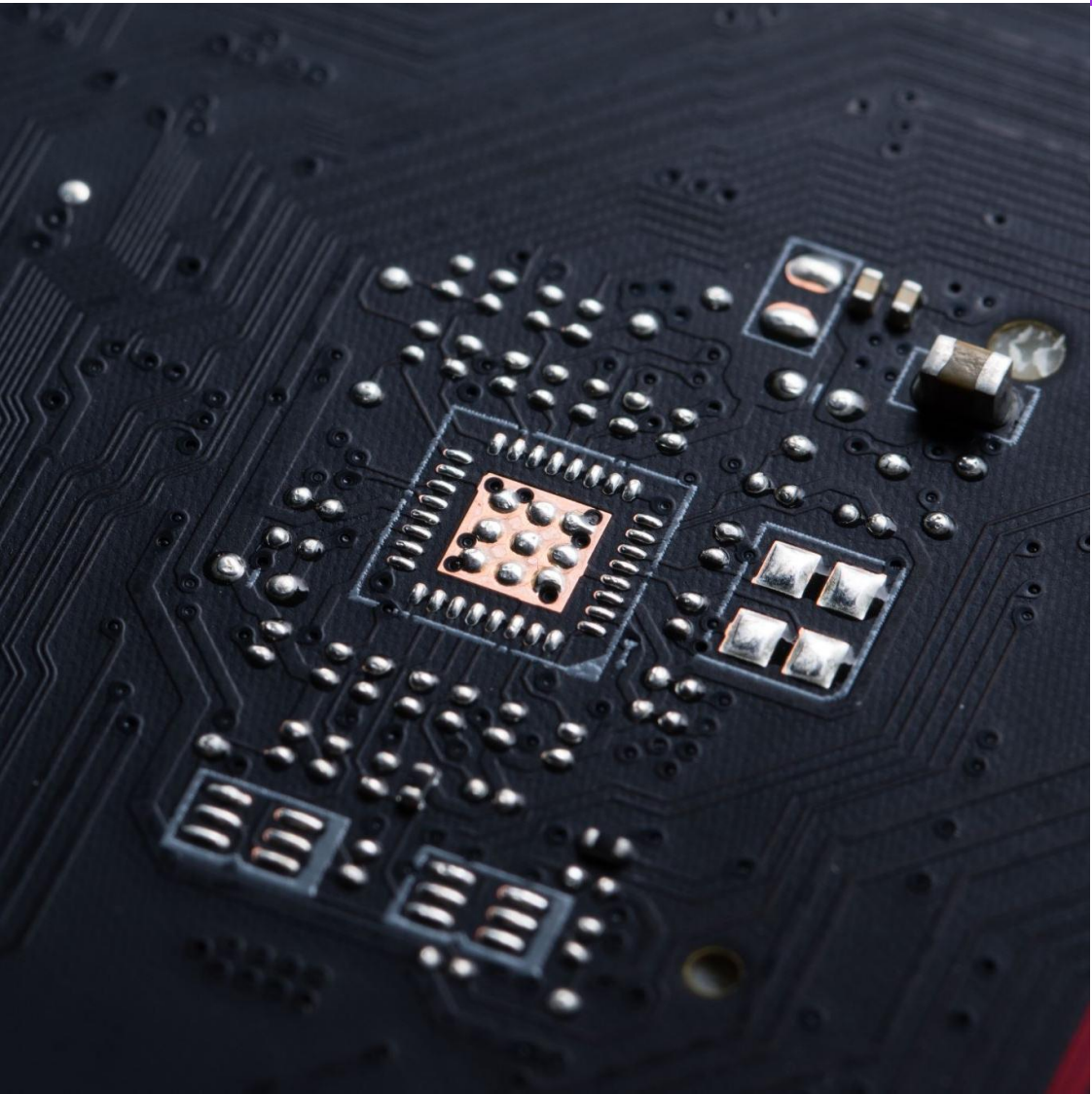
The Case

Peripherals – Mouse/Keyboard/Monitor/Speakers

Bonus – RGB

- + Cables
- + GPU
- + Ram
- + Mobo
- + Fans
- + Cooling
- + The case





ASSEMBLY

- + RAM DIMM order
- + Thermal Paste
- + PCIE

References

- + <https://pcpartpicker.com/>
- + <https://www.newegg.com/>
- + <https://versus.com/en>
- + https://en.wikipedia.org/wiki/Moore%27s_law
- + https://en.wikipedia.org/wiki/Central_processing_unit
- + <https://www.youtube.com/@LinusTechTips>
- + <https://www.youtube.com/@Jayztwocents>